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Title of the article: *Detection of American Football Head Impacts Using Biomechanical Features and Support Vector*

The content extracted is as follows:

implemented on a wearable sensor for detecting field football head impacts.

Our method incorporates a support vector machine classifier that uses biomechanical features from the time domain and frequency domain, as well as model predictions of head-neck motions.

The classifier was trained and validated Accumulation of head impacts may contribute to acute and long term brain trauma.

In rodent models, repeated head impacts can worsen brain injury outcomes and substantially prolong recovery^{1 3}.

Even without clinically diagnosed concussions, players with a greater number of head impacts in a game showed higher levels of blood brain barrier disruption than those with fewer impacts⁶.

At lower impact severities and exposure levels, a bout of soccer headers led to acute neurological deficits and concussion symptoms^{7,8}.

Aside from acute injuries, head impacts over a season of football has been shown to lead to white matter integrity changes⁹.

In fact, long-term head impact exposure from sports and other activities may trigger a neurodegenerative condition called chronic traumatic encephalopathy^{10,11}.

With heightened awareness of the association between sports head impacts and brain trauma, guidelines have been introduced to limit head impact exposure in sports, such as the recent US Soccer organization's recommendation of no heading for young children¹².

In response to such concern, many head impact sensors have been developed to measure head impact exposure.

Some sensors use simple linear acceleration thresholds to differentiate head impacts from normal motions.

Therefore, employing more features from the time and frequency domains of kinematics as well as machine learning algorithms may be necessary to generate a high-performance classifier.

In detecting head impacts, one study used a neural network algorithm combined with discrete Fourier transform heuristics to distinguish between head impacts and nonimpacts in soccer³¹.

Raw time-domain data were used as features in the neural network classifier, and the classifier was evaluated to have 88% sensitivity and 47% specificity.

In pilot work³², we have demonstrated efficacy of power spectral density (PSD) features combined with an SVM classifier in differentiating dummy head impacts and laboratory-simulated nonimpacts with 98% sensitivity and 99.98% precision.

In the current work, our objective was to identify the distinguishing features of collegiate football head impacts in the field, and use these features to develop and validate a head impact classifier.

Methods

We deployed instrumented mouthguard sensors to collegiate football players during regular season games and practices to gather a ground truth dataset of head impacts and nonimpacts to train and validate a head impact classifier (Fig. 1).

In the training and validation set, we included video-verified helmet contact recordings as head impacts, and video-verified noncontact recordings as nonimpacts.

Then, features were extracted from the kinematic sensor measurements to train an SVM classifier that distinguishes between impacts and nonimpacts.

The number of players and events were chosen to acquire a similarly sized dataset as the previous laboratory impact detection study using dummy head impacts and human subject reconstructed nonimpacts ($n_{\text{impact}} = 128$, $n_{\text{nonimpact_hiIR}} = 118$)³².

Since the application of interest was head impact detection, we were most interested in distinguishing data recorded during head impacts (helmet impacts in the case of football) from other mouthguard recordings (e.g. running, jumping, mouthguard manipulation without contact).

The system was not trained on head acceleration activity without direct head contact (e.g. where body contact could lead to head acceleration without direct head impact), since it is difficult to judge and label such activities from video.

Body contacts that may have involved head accelerations were excluded from the head impact classifier training for this study.

Since the mouthguard timestamp had a resolution of 1 second, and video-mouthguard synchronization could also have at least ± 1 second error, we used a ± 2 second time window to select potential helmet contact periods from the mouthguard data, i.e. any mouthguard recording occurring within 2 seconds of a video-identified helmet contact period was selected. Second, to further ensure that the mouthguard helmet contact recordings matched with video-observed helmet contacts, an additional helmet impact directionality constraint was imposed where we eliminated impacts for which neither the peak acceleration direction nor the integrated head motion direction matched with the video-observed impact directionality.

For feature extraction, raw kinematic signals gathered from the mouthguard (triaxial linear acceleration and angular velocity time series) were rotated to the head anatomical frame, but the linear accelerations were not transformed to the centre of gravity of the head, to avoid cross-coupling of linear and angular features.

Although the features extracted from the kinematics signals used untransformed linear accelerations to avoid cross-coupling of linear and angular features, the model-based features used linear accelerations transformed to the centre of mass of the head to estimate head velocities and displacements.

A two-class SVM classifier was trained to differentiate head impacts from the nonimpacts.

The off-teeth recordings were not included in the SVM classifier training, since in practice we would be directly rejecting them using the IR feature.

First, sequential feature selection was used to prune the feature set to avoid overfitting (i.e. when the number of training examples is small compared to the number of fitting parameters, such that the fit is good for the training data but is not generalizable) and optimize classifier performance, using the MATLAB `sequentialfs` function.

The feature selection used cost functions to optimize the classifier for (1) area under the receiver operating characteristics (ROC) curve (AUC), and (2) the F-measure, which is the harmonic mean of sensitivity and precision.

In the leave-one-out process, we trained the classifier with all but one sample from the full dataset (including impacts and nonimpacts), then used the trained classifier to classify the remaining sample, and aggregated the results from all possible combinations to calculate the cost function (i.e. number of cycles equals size of training/validation dataset).

Second, after feature selection, the leave-one-out cross validation performance of the AUC-optimized and F-measure-optimized classifiers was used to compare the two classifiers, specifically using sensitivity, specificity, accuracy, precision, area under the ROC curve, and area under the precision-recall (PR) curve as the performance criteria.

To compare the performance of the optimized classifiers against a basic method, we also plotted the ROC and PR curves for a simple acceleration thresholding method, where the peak linear acceleration vector magnitude was used to classify between impacts and nonimpacts.

where true positive (TP) was a head impact that was classified by the classifier as a head impact, true negative (TN) was a nonimpact that was classified by the classifier as a nonimpact, a false positive (FP) was a nonimpact that was classified by the classifier as a head impact, and a false negative (FN) was a head impact that was classified by the classifier as a nonimpact.

Results

On the other hand, a simple classifier just based on linear acceleration magnitude

thresholding (red) did not perform much better than randomly assigning classes with a probability of impact of 50% (with area under the ROC curve close to 0.5).

The independent youth test dataset included 16 head impacts and 16 nonimpacts.

for a college offensive lineman, 46 of the 1219 data recordings were classified as head impacts, and the remaining recordings were either filtered out using IR (n = 1026) or classified as nonimpacts (n = 147).

Shown in Fig. 8 are time points of classified head impacts compared with observations from video analysis.

Over the recorded video duration, we show video-identified periods where helmet or body contacts were observed and periods where contacts may have occurred but view of the player's head was obstructed. Most of the classified head impacts fell within or close to periods of observed or possible contact based on video analysis.

Discussion

Here we present a study where we trained and validated a head impact classifier with collegiate field data.

We combined the use of a head impact sensor with independent video analysis to provide training and validation data from collegiate football for the head impact classifier.

Using this methodology, we have identified features that were helpful in distinguishing between helmeted football head impacts and nonimpacts on the collegiate football field.

Extending to other wearable head impact sensors, we advise that sensors include a suitable proximity sensing mechanism to sense device placement and reject recordings when the device is not properly mounted. The SVM classifier helped to reject high-IR nonimpacts.

Some helmet contact recordings identified with the classifier were in periods with obstructed view of the player's helmet during contact, indicating that the impact detection algorithm may help to more sensitively identify helmet contact when the player's action is obstructed in video.

Traditional peak kinematics features often used for impact detection could not distinguish between head impacts and nonimpacts as well as frequency-domain features.

As seen in Fig. 3, the peak kinematics values largely overlapped between impact and nonimpact distributions, resulting in low predictive value for head impact detection, as confirmed by the red ROC and PR curves in Fig. 7.

On the other hand, head impacts involve more massive collisions between players and may thus exhibit lower frequency dynamics with higher power.

The classifier is only evaluated to be used with the specific instrumented mouthguard design used in this study for classifying collegiate American football helmet impacts.

Retraining is likely required for any device that does not have the same mechanical and electrical design as this instrumented mouthguard and for detecting head impacts in other contexts.

Although features of helmet impacts recorded on different head impact sensors may be similar, provided that these sensors can accurately measure skull motion, the kinematics of nonimpact recordings will likely differ for different device form factors and sensor locations.

It is also possible that different helmet materials may lead to different head impact kinematics.

In unhelmeted sports or other sports where different forms of contact and/or player behaviours are involved, impact and nonimpact characteristics are expected to change, requiring retraining and validation of the classifier.

Prior to implementing the impact detection system for field use independent of video confirmation, testing of the classifier using an independent collegiate dataset from different players and positions will help further validate field performance.

For example, almost all recordings from when the player was buried under a pile were excluded from training, since a clear view of the player's head/helmet could not be established. However, it is highly likely that helmet contact will occur during such periods and it may exhibit unique dynamics due to multiple impacts and simultaneous impacts to different objects (body, helmet, ground).

In the context of head impact detection, although high kinematic accuracy of sensors is not required, such noise may make it more difficult to differentiate between impacts and nonimpacts.

For example, if the mouthguard is not coupled to the teeth during head impacts, even though the impact detection algorithm may be able to reject such recordings, it is not ideal since we fail to capture the head impact data.

It is also possible that some head impacts may have been missed due to poor coupling of the mouthguard with the upper jaw during impact from deliberate removal or loose fitting.

On the collegiate football field, we found that nonimpacts could account for most recordings on a small light-weight wearable head impact sensor.

Typically, for an offensive lineman involved in most plays over a game, we can expect the instrumented mouthguard to record up to 50 head impacts and about 1000 nonimpacts (that are mostly off-teeth recordings).

It is important in this context that classifiers be evaluated for sensitivity and precision, instead of sensitivity and specificity, since even 90% specificity would lead to 100 false positive detections, which is twice the number of actual head impacts.

Given over 90% sensitivity and precision, up to 5 head impacts may be missed and 5 detected head impacts may be false positives.

To collect a dataset for studying subconcussive head impacts, this level of sensitivity should be sufficient to allow for gathering of a large amount of data in a relatively short amount of time, since 90% of the helmet contacts can be captured. However, to gather concussion data, which are rare occurrences, higher sensitivity may be desirable.

Future wearable head impact sensor designs should focus on (1) optimizing skull coupling and minimizing noise in the head impacts, such that these events may be more clearly distinguished from nonimpacts, and (2) ensuring player compliance such that the device is worn and well-coupled during all potential contact activities.